JOSEPH MOSKOVITZ

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PERSONAL STATEMENT

Aerospace Engineering BS/MS candidate specializing in electric propulsion and nuclear technology, with a strong foundation in M&A finance and space policy. Experienced in international and collaborative academic research across multiple international teams. Passionate about driving the long-term technological development and commercialization of nuclear power and emerging technologies within the space domain.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY — Atlanta, GA

Master's in Science in Aerospace Engineering — Aug 2026 - May 2027

Cumulative GPA: 4.0/4.0

Specialized Coursework: Heat Transfer, Radiation Transport, Nuclear Materials

Bachelor's in Science in Aerospace Engineering — Aug 2022 - May 2026

Minor in Nuclear & Radiological Engineering

Cumulative GPA: 4.0/4.0 | Dean's List 2022-2026

Specialized Coursework: Reactor Physics, Plasma Physics & Fusion Engineering, Radiation Protection Engineering, Space Policy, Finite Element Analysis, Scientific Machine Learning

RANSOM EVERGLADES SCHOOL — Miami, FL

High School Diploma — Aug 2018 - May 2022

Cumulative GPA: 4.0/4.0 | Cum Laude

RESEARCH EXPERIENCE

HIGH-POWER ELECTRIC PROPULSION LAB - GEORGIA TECH — Atlanta, GA

Undergraduate Research Assistant — Jan 2024 – current

- **Machine Learning (ML) to Predict Low-Power HET Performance Operating on Multiple Propellants**
  - Collaborating with an international team (US, India, Israel) to develop a HET data preprocessing pipeline and evaluate various ML models predicting output parameters across various propellants (Xe, Kr, Ar, CO_2, N_2).
  - *Note: accepted to AIAA SciTech 2027 and under peer review at the Journal of Electric Propulsion*
- **Comprehensive Survey of Electric Propulsion (EP) Missions [2017-2025]**
  - Developed python-based web scrapers and automation scripts to extract and aggregate comprehensive mission data from ESA's DISCOS and Space-Track.org databases.
  - Analyzed orbital data to evaluate and quantify industry-wide adoption trends of EP technologies over the past decade.
- **Retarding Potential Analyzer (RPA) Redesign**
  - Redesigned the lab's proprietary RPA housing, grids, and wiring to more accurately analyze different plasmas and prevent shorting of different potential grids on older models.
  - Rebuilt CAD assemblies in SolidWorks, generating part drawings and GD&T with rigorous Design for Manufacturing (DFM) considerations for small, fragile components.
  - Created a BOM for the RPA, talked with various manufacturers and received quotes for different parts, machined parts in-house and ordered parts externally via lab management & procurement platform, and finally assembled 6+ fully functional units with interchangeable & spare parts.

EQUIPO DE PROPULSIÓN ESPACIAL Y PLASMAS - UC3M — Madrid, ES

Undergraduate Research Assistant — Sep 2024 – Dec 2024

- Developed a centralized machine/computer controls interface with LabVIEW for all power supply units (PSUs) involved in operating a HET.

ELECTRIC PROPULSION LABORATORY - TECHNION — Haifa, IL

Undergraduate Researcher — Jun 2024 – Aug 2024

- Characterized the performance and efficiency of the Simplified CAMILA low-power Hall-Effect Thruster (HET) across a diverse set of monatomic and molecular propellants (Xe, Kr, Ar, CO_2, N_2).
- Evaluated the viability of alternative propellants to mitigate industry reliance on Xe.
- *Note: Findings published at IEPC 2025 and at the Journal of Electric Propulsion.*

WORK EXPERIENCE

REALTA FUSION Madison, WI
Fusion Intern May 2026 – Aug 2026

- Developed a framework to cross-validate neutral beam injection (NBI) beam divergence from multiple existing diagnostics to better quantify beam characteristics.
- Designed and coordinated an experimental campaign with results to be presented at the American Physical Society Depart of Plasma Physics (APS DPP) conference in November.

ALLEN & COMPANY New York City, NY
Investment Banking Summer Analyst Jun 2025 – Jul 2025

- Executed live deals alongside senior bankers across the Biopharma, SaaS, and sports industrial verticals.
- Leveraged comprehensive valuation techniques (DCF, comps, precedents), M&A analysis (accretion/dilution), and rigorous equities research to support transaction advisory.

GEORGIA TECH COLLEGE OF ENGINEERING Atlanta, GA
Deformable Bodies Teaching Assistant Jan 2024 – May 2024

- Developed homework problems, facilitated office hours, and graded homework and exams for GT COE 3001, ensuring comprehension of complex engineering mechanics.

PUBLICATIONS

J. Moskovitz, W. Brabston, D. Lev, M. L. R. Walker, M. Rubanovych, and V. Balabanov. "Performance comparison and analysis of low-power hall thruster operation on atomic and molecular propellants." *J Electric Propulsion* **5**, 32 (2026). https://doi.org/10.1007/s44205-026-00199-5

PROJECTS

SCRAP-SAT Atlanta, GA
Magnetics Engineer & Finance Lead Jan 2026 – current

- Developing an integrated satellite to detumble and deorbit uncooperative space debris in Low-Earth Orbit (LEO) using magnetic eddy braking and a GOLD passive drag package.
- Developed a Python-based design tool using the magpylib library to optimize High-Temperature Superconductor (YBCO) pancake coils; performed multi-variable parameter sweeps (radius, turns, stacking) to maximize magnetic field strength while adhering to geometric constraints.
- *Note: white paper accepted to AIAA SciTech 2027.*

1-D REACTION WHEEL Atlanta, GA
Project Lead Jan 2026 – May 2026

- Directing the design and construction of a reaction/momentum wheel to demonstrate spacecraft attitude control systems.
- Manufacturing the base, demo satellite body, and wheel via 3D printing while sourcing commercial-off-the-shelf electronic and mechanical components.
- Modeling open and closed-loop system behavior in Simulink and designing a PID controller for precision attitude control.
- Constructing circuits to power the stepper motor, Arduino controller, and IMU locally on the satellite disk.

COMMUNITY LEADERSHIP

CHABAD ON CAMPUS Atlanta, GA
Executive Board Member Jan 2024 – current

- Plan holidays, parties, social events, and more for the Jewish community at Georgia Tech.
- Increased the number of events for each Jewish holiday and helped increase weekly Shabbat attendance by over 50% in 3 years.

ZETA CHAPTER OF ALPHA EPSILON PI Atlanta, GA
Exchequer Apr 2023 – Apr 2025

- Managed the finances of the fraternity, including collecting dues, filing taxes, paying bills, and setting the budget.
- Incorporated the Chapter as a 501(c)(7) nonprofit organization, acquired a corporate line of credit, implemented modern expense-tracking software, and authored the organization's official Expense Policy and had it passed into the bylaws.

ADDITIONAL

Technical Skills: SolidWorks, Python, MATLAB, Simulink, LabVIEW, Finite Element Method Magnetics (FEMM), Data Analysis, Machine Learning, Excel, Woodworking, 3D Printing, Circuit design & construction
Languages: Bilingual in English & Spanish, Conversational in Hebrew